Afaq Virk

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Education

Carleton University *Sep 2024–Dec 2026*
B.Comp.Sc. (Honours), Industrial Applications Stream Ottawa, ON
Minor in Economics | **CGPA: 11.9/12.0**

Work Experience

Shopify *Aug 2024–Present*
Full-Stack + Applied ML Engineering Intern Ottawa, ON

- Selected as **1 of 30 high school students internationally** for Shopify's Dev Degree program, receiving full tuition sponsorship and a 3–4-year internship across teams including Analytics and Capital Agents.
- Built foundational evaluation pipelines for fine-tuned text-to-SQL LLM models used on Shopify's Analytics page.
- Built ML tooling connecting Shopify's banking, financing, and reconciliation tools to Sidekick AI; implemented referral attribution for an AI channel associated with US$1.9M in merchant financing.
- Hardened Python/Airflow analytics pipelines with ClickHouse disk spilling, bounded concurrency, and StatsD monitoring as merchant coverage expanded 9.8×, from 45,000 to 442,000 stores.
- Built Airflow DAGs and Apache Flink pipelines for RAG to help merchants discover shop insights; improved multilingual rendering for an analytics product reaching 165M impressions across 490,000+ stores.
- Re-evaluated several existing AI workflows, reducing tool costs by approximately 50% and exposing 5× cost regressions in web searches.

MathemaTech *May 2021–Present*
Founder, Lead Educator Remote

- Secured a $3,000 Ontario Ministry grant to launch the business, grew it to $3,000 MRR, and delivered 3,800+ hours of one-on-one mathematics, computer science, and physics tutoring to 150+ secondary and university students in Canada and internationally.
- Created and marketed educational content reaching 700,000+ students and educators and generating 2M+ monthly social impressions; building AI tools to automate practice generation and student tracking.

Upper Canada District School Board *Aug 2023–Jul 2024*
Member, Board of Trustees (Student Trustee) Brockville, ON

- Elected to represent 27,000+ students; led board discussions on STEM, classroom AI, and pedagogy modernization and oversaw $6,000+ in fundraising across 21 schools for CHEO.
- Organized three annual conferences with approximately $75,000 average budgets for about 80 trustees from 35+ school boards and coordinated senior stakeholders, including Ontario's Minister of Education.

Selected Projects

Donair Feb 2026
TypeScript, Next.js, Supabase (Postgres + pgvector), Gemini, Solana Pay

- Built an agentic crowdfunding platform with semantic donor discovery across 110,000+ historical records, weighted capacity, affinity, and timeliness scoring, and real-time Solana Pay flows; secured $8,000 in funding from Carleton University through the Nicol Ventures platform and won the Talsom Sponsor Challenge and Front Row Ventures Investor Pitch Award.

LoRaWATsGoingOn Jan 2026
Python, FastAPI, SQLite, LoRaWAN (ChirpStack)

- Built a ChirpStack-powered device monitor with device-health analytics, anomaly annotations, and gateway navigation; won 1st place at the SSi Canada x Carleton Computer Networks Hackathon.

K.A.R.E.N Jan 2026
SystemVerilog (FPGA), Tcl, Python, Vivado; Gemini; ElevenLabs

- Built an FPGA assistant with HDMI video and keyboard I/O, a constrained LLM-to-hardware command vocabulary for safe deterministic execution, and synchronized voice input, voice cues, and waveform display.

elECHOlocation Oct 2023
Arduino, C++, ultrasonic sensors

- Engineered and experimentally evaluated an ultrasonic navigation aid for visually impaired users; earned $6,000+ in awards, including Gold, Outstanding Research, and the CAP Physics Prize, and represented Canada at MILSET Mexico.

Skills & Technologies

Languages: Rust, Python, C, C++, C#, Java, TypeScript, SQL, SystemVerilog
Frameworks: React, Next.js, FastAPI, Rails, Apache Flink, LangChain, DSPy
Data/Infra: PostgreSQL (pgvector), BigQuery, Airflow, REST APIs, WebSockets
ML/AI: LLM evaluation, RAG, agent design, inference orchestration
Systems/Hardware: LoRaWAN, embedded systems, FPGA digital logic, HDMI/video timing
Tools: Git, Docker, Linux, GCP, Vivado/Tcl, Blender